AAKRITI KARNA

+91-8092161091 AkreetyKarna123@gmail.com LinkedIn GitHub

Professional Summary

Early career developer who enjoys breaking down complex problems into clean, practical solutions and turning ideas into functional, reliable software.

Education

Vel Tech University, Chennai 2022 – 2026
Bachelor of Technology in Computer Science and Engineering CGPA: 7.3

Technical Skills

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Languages: Python, Java, JavaScript, SQL
Backend: Django, Node.js, Express.js, REST APIs, JWT Authentication
Databases: PostgreSQL, MySQL, MongoDB
Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3
Tools: Git, GitHub, VS Code, CI/CD Basics

Experience

Frontend Developer Intern — Sayathari Tech, Kathmandu, Nepal (On-site) *Jun 2026 – Sep 2026*

- Partnered with backend developers to integrate RESTful APIs, resolving recurring data synchronization issues between frontend and backend services.
- Engineered and maintained responsive, cross-browser user interfaces using HTML5, CSS3, and JavaScript across multiple device and browser combinations.
- Debugged and resolved 15+ UI issues across browsers and screen sizes, improving application stability and user experience.
- Streamlined page load performance and optimized component rendering to reduce client-side rendering time.

Projects

Inventory Management System *2026*
React 19, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, JWT Authentication

- Built a full-stack inventory management dashboard using React 19, Vite, and Tailwind CSS on the frontend, with Node.js and Express.js powering the backend.
- Designed RESTful CRUD APIs connected to MongoDB, handling core inventory operations like adding, updating, and tracking stock.
- Implemented JWT-based authentication and bcrypt password hashing to secure user accounts and protect API routes.
- Built analytics dashboards with Recharts to visualize stock trends and key inventory metrics for users.

Train Delay Prediction API System *Dec 2025 – Apr 2026*
Node.js, Express.js, MongoDB, Socket.io, JWT Authentication

- Built 12+ RESTful API endpoints using Node.js and Express.js to manage train and delay-related data.
- Designed the MongoDB schema to structure and efficiently query train schedules and delay records.
- Implemented JWT authentication and bcrypt password hashing to secure API access and protect user data.
- Integrated Socket.io to push real-time delay notifications and live status updates to connected clients.

User-Based Collaborative Filtering – Movie Recommender System *2026*
Python, Scikit-learn / Surprise Library, MongoDB, Django

- Built a movie recommendation system in Python using the Surprise library, applying user-based collaborative filtering on the MovieLens dataset.
- Computed user-user similarity from historical rating patterns to predict how a user would rate unseen movies.
- Generated top-N personalized movie recommendations for each user based on predicted ratings from similar users.
- Developed a Django web interface allowing users to view their ratings and browse personalized recommendations.

Certifications

- Microsoft SQL Learning Path – Microsoft Learn
- Oracle Java Foundations – Oracle
- GitHub Foundations – GitHub

Achievements & Leadership

- Secured a **Top-3 position** in a 24-hour hackathon by developing and presenting a working backend solution in a competitive team environment.
- Authored and presented research paper *"AI-Driven Delay Forecasting Using CatBoost for Congested Railway Sections"* at the 4th International Conference on Electronics and Renewable Systems (ICEARS 2026).
- Contributed to open-source projects and helped organize college technical events, collaborating with student developer communities.
- Strengthening **Data Structures & Algorithms** skills through LeetCode, focusing on Arrays, Strings, and Hashing.